


CLOSED-FORM SOLUTION (for linear models)

- Loss:

$$L(X, y, \underline{w}) = \frac{1}{n} \|y - X\underline{w}\|^2$$

- min Loss $\Rightarrow$ find where it is $= 0$

$$\frac{\partial L}{\partial \underline{w}} = 0 \rightarrow \frac{2}{n} \cdot (y - X\underline{w})^T \cdot X = 0$$

- To get $\frac{2}{n}$:

$$(y - X\underline{w})^T X = 0$$

- Apply the property: $(A - B)^T = A^T - B^T$

$$[y^T - (X\underline{w})^T] \cdot X = 0$$

- Apply the distributive property of products:

$$(a - b) * c = a * c - b * c$$

$$y^T X - (X\underline{w})^T X = 0$$

- Applico la p.t.a: $(AB)^T = B^T \cdot A^T$

$$y^T X - w^T X^T X = 0$$

- Porto a dx $w^T X^T X$:

$$y^T X = w^T X^T X$$

- Esplicito il valore di w
dividendo entrambi i termini
per $(X^T X)$

$$w^T = (X^T X)^{-1} y^T X$$

TRAINING with MINIBATCH SGD:

1. $\underline{w}$ initialized randomly
(INCLUDE THE BIAS)

2. while CONVERGENCE:

2.1] β = random subset of
the training set

$$2.2] \underline{w} \leftarrow \underline{w} - \frac{\eta}{|\beta|} \cdot \sum_{i \in \beta} \partial_{\underline{w}} \ell^{(i)}(\underline{w}) \quad (*)$$

$$\underline{w} \leftarrow \underline{w} - \frac{\eta}{|\beta|} \cdot \sum_{i \in \beta} (\underline{w} \underline{x} - y^{(i)}) \cdot \underline{x}$$

*) ANALITICALLY:

$$\begin{aligned} \ell^{(i)}(\underline{w}) &\stackrel{\text{DEF LOSS}}{=} \frac{1}{2} (\hat{y}^{(i)} - y^{(i)})^2 = \\ &= \frac{1}{2} (\underline{w} \underline{x} - y^{(i)})^2 \end{aligned}$$

$$\frac{\partial \ell^{(i)}}{\partial \underline{w}} = \frac{1}{2} \cdot 2 (\underline{w} \underline{x} - y^{(i)}) \cdot \underline{x}$$

$\forall \underline{x} \in \beta$

Loss Function

for CLASSIFICATION.

LOSS SU
UN SINGOLO
↓ CAMPIONE

$$l = - \sum_{j=1}^q y_j \cdot \log(\hat{y}_j)$$

q : nb. of classes
|||
nb. of output
neurons

• ESPANDO LA SOFTMAX: $\hat{y}_j = \frac{\exp(o_j)}{\sum_{i=1}^q \exp(o_i)}$

$$l = - \sum_{j=1}^q y_j \cdot \log\left(\frac{\exp(o_j)}{\sum_{i=1}^q \exp(o_i)}\right)$$

• APPLICO $\log\left(\frac{a}{b}\right) = \log a - \log b$

$$L = - \sum_{j=1}^q y_j \cdot \left(\overset{\exp(o_j)}{o_j} - \log\left(\sum_{i=1}^q \exp(o_i)\right) \right)$$

• DISTRIBUIAMO IL PRODOTTO:

$$L = - \sum_{j=1}^q y_j \cdot o_j + \sum_{j=1}^q y_j \cdot \log\left(\sum_{i=1}^q \exp(o_i)\right)$$

- R I P O R T O D A S O P R A

$$l = - \sum_{j=1}^9 y_j \cdot o_j + \underbrace{\sum_{j=1}^9 y_j \cdot \log \left(\sum_{i=1}^9 \exp(o_i) \right)}_{\text{VALE 1 solo per 1 valore}}$$

$$= - \sum_{j=1}^9 y_j o_j + \log \left(\sum_{i=1}^9 \exp(o_i) \right)$$

- Derivo la loss rispetto a o_j :

$$\frac{\partial l}{\partial o_j} = -y_j + \frac{\partial}{\partial o_j} \left[\log \left(\sum_{i=1}^9 \exp(o_i) \right) \right]$$

- Applico $\frac{\partial}{\partial x} [\log(f(x))] = \frac{1}{f(x)} \cdot f'(x)$

$$= -y_j + \frac{\exp(o_j)}{\sum_{i=1}^9 \exp(o_i)} =$$

$\frac{\partial}{\partial x} \log(f(x))$

$$= \text{softmax}(o_j) - y_j$$

→ MISURA
LA
DISCREPANZA

$$\frac{\partial (e^x)}{\partial x} = e^x$$